

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 6E nr. 11.1

1. $f(x) = \cosh x + \cos x - \frac{x^4}{12}$

2. $f'(x) = \sinh x - \sin x - \frac{x^3}{3} \Rightarrow f'(0) = 0$

3. $f''(x) = \cosh x - \cos x - x^2 \Rightarrow f''(0) = 1 - 1 - 0 = 0$

4. $f^{(3)}(x) = \sinh x + \sin x - 2x \Rightarrow f^{(3)}(0) = 0$

5. $f^{(4)}(x) = \cosh x + \cos x - 2 \Rightarrow f^{(4)}(0) = 1 + 1 - 2 = 0$

6. $f^{(5)}(x) = \sinh x - \sin x \Rightarrow f^{(5)}(0) = 0$

7. $f^{(6)}(x) = \cosh x - \cos x \Rightarrow f^{(6)}(0) = 1 - 1 = 0$

8. $f^{(7)}(x) = \sinh x + \sin x \Rightarrow f^{(7)}(0) = 0$

9. $f^{(8)}(x) = \cosh x + \cos x \Rightarrow f^{(8)}(0) = 1 + 1 = 2$

$\Rightarrow f^{(k)}(0) = 0$ voor $k = 1, \dots, 7$, en $f^{(8)}(0) = 2 > 0$, dus lokaal minimum in $x = 0$.

References